MongoDB Lab Exercises

Database: sample_mflix

Cluster: cluster0.5qruewy.mongodb.net

User: tusharverma_db_user

This document contains MongoDB find, update, aggregate, and CRUD queries executed against the sample_mflix dataset, plus a custom collection my_movies_tushar. Screenshots from MongoDB Compass are attached for selected tasks.

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Tasks 1-7 - Find / count / filter / sort

Tasks 8-10 - Insert & update (my_movies_tushar)

Tasks 11-17 - Advanced queries ($in, $all, $regex, skip/limit)

Tasks 18-20 - updateMany / $push / deleteMany

Tasks 21-23 - Aggregations ($unwind, $group, $lookup)

Tasks 24-26 - Pending

Tasks 27-28 - Director & country aggregations

#1 Find one movie document

```
db.movies.findOne({})
```

#2 Movies from year 2000 (title & year only)

```
db.movies.find(
    { year: 2000 },
    { title: 1, year: 1, _id: 0 }
)
```

#3 Count PG-13 rated movies

```
db.movies.countDocuments({ rated: "PG-13" })
```

#4 Movies with runtime greater than 180

```
db.movies.find({
    runtime: { $gt: 180 }
})
```

#5 Top 5 longest movies (numeric runtime)

```
db.movies.find({
    runtime: { $type: "number" }
}).sort({ runtime: -1 }).limit(5)
```

#6 Movies with genre Comedy

```
db.movies.find({
    genres: "Comedy"
})
```

#7 Movies featuring Tom Hanks (find + count)

```
db.movies.find({
    cast: "Tom Hanks"
})

db.movies.countDocuments({
    cast: "Tom Hanks"
})
```

#8 Insert one document into my_movies_tushar

```
db.my_movies_tushar.insertOne({
  title: "The Dark Knight",
  year: 2008,
  runtime: 148,
  genres: ["Action", "Sci-Fi"],
  rated: "PG-13",
  directors: ["Christopher Nolan"]
})
```

#9 Insert many documents into my_movies_tushar

```
db.my_movies_tushar.insertMany([
  {
    title: "Inception",
    year: 2010,
    runtime: 152,
    genres: ["Action", "Sci-Fi"],
    rated: "PG-13",
    directors: ["Christopher Nolan"]
  },
  {
    title: "Dhurandhar",
    year: 2010,
    runtime: 200,
    genres: ["Action", "Crime", "Drama"],
    rated: "A",
    directors: ["Christopher Nolan"]
  },
  {
    title: "Interstellar",
    year: 2010,
    runtime: 180,
    genres: ["Action", "Sci-Fi", "Romance"],
    rated: "PG-13",
    directors: ["Christopher Nolan"]
  }
])
```

#10 Find and update Dhurandhar (rated + directors)

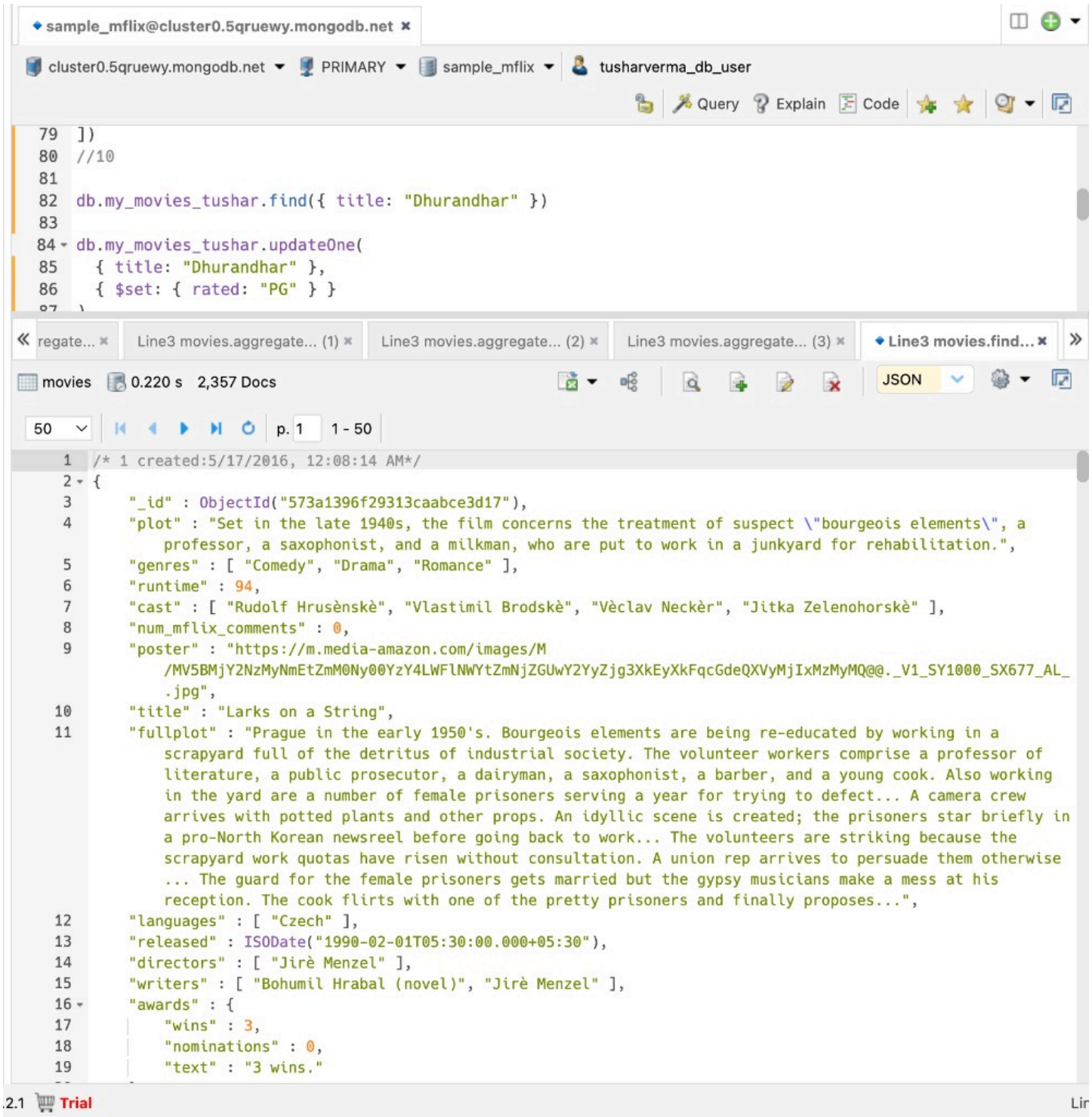
```
db.my_movies_tushar.find({ title: "Dhurandhar" })

db.my_movies_tushar.updateOne(
  { title: "Dhurandhar" },
  { $set: { rated: "PG" } }
)

db.my_movies_tushar.updateOne(
  { title: "Dhurandhar" },
  { $set: { directors: "Aditya Dhar" } }
)

db.my_movies_tushar.find({ title: "Dhurandhar" })
```

Screenshot - Task 10 (MongoDB Compass)



#11 Drama/Romance movies from 1990-1999

```
db.movies.find({
  genres: { $in: ["Drama", "Romance"] },
  year: { $gte: 1990, $lte: 1999 }
})
```

#12 Movies that are both Action and Sci-Fi

```
db.movies.find({
  genres: { $all: ["Action", "Sci-Fi"] }
})
```

#13 High-rated movies with many IMDb votes

```
db.movies.find(
  {
    "imdb.rating": { $gte: 8.5 },
    "imdb.votes": { $gt: 100000 }
  },
  { title: 1, "imdb.rating": 1, _id: 0 }
).sort({ "imdb.rating": -1 })
```

#14 Titles starting with 'The Lord'

```
db.movies.find({
  title: { $regex: "^The Lord" }
})
```

#15 Movies missing the rated field

```
db.movies.find({
  rated: { $exists: false }
})
```

#16 Count Indian movies after 2010

```
db.movies.countDocuments({
  countries: "India",
  year: { $gt: 2010 }
})
```

#17 Comedy movies - sorted, skip 20, limit 10

```
db.movies.find({ genres: "Comedy" })
  .sort({ title: 1 })
  .skip(20)
  .limit(10)
```

#18 Increment views on all my_movies_tushar docs

```
db.my_movies_tushar.find()

db.my_movies_tushar.updateMany({}, { $inc: { views: 1 } })
```

#19 Push Thriller genre onto Dhurandhar

```
db.my_movies_tushar.find({ title: "Dhurandhar" })

db.my_movies_tushar.updateOne(
  { title: "Dhurandhar" },
  { $push: { genres: "Thriller" } }
)
```

#20 Delete all PG-rated docs from my_movies_tushar

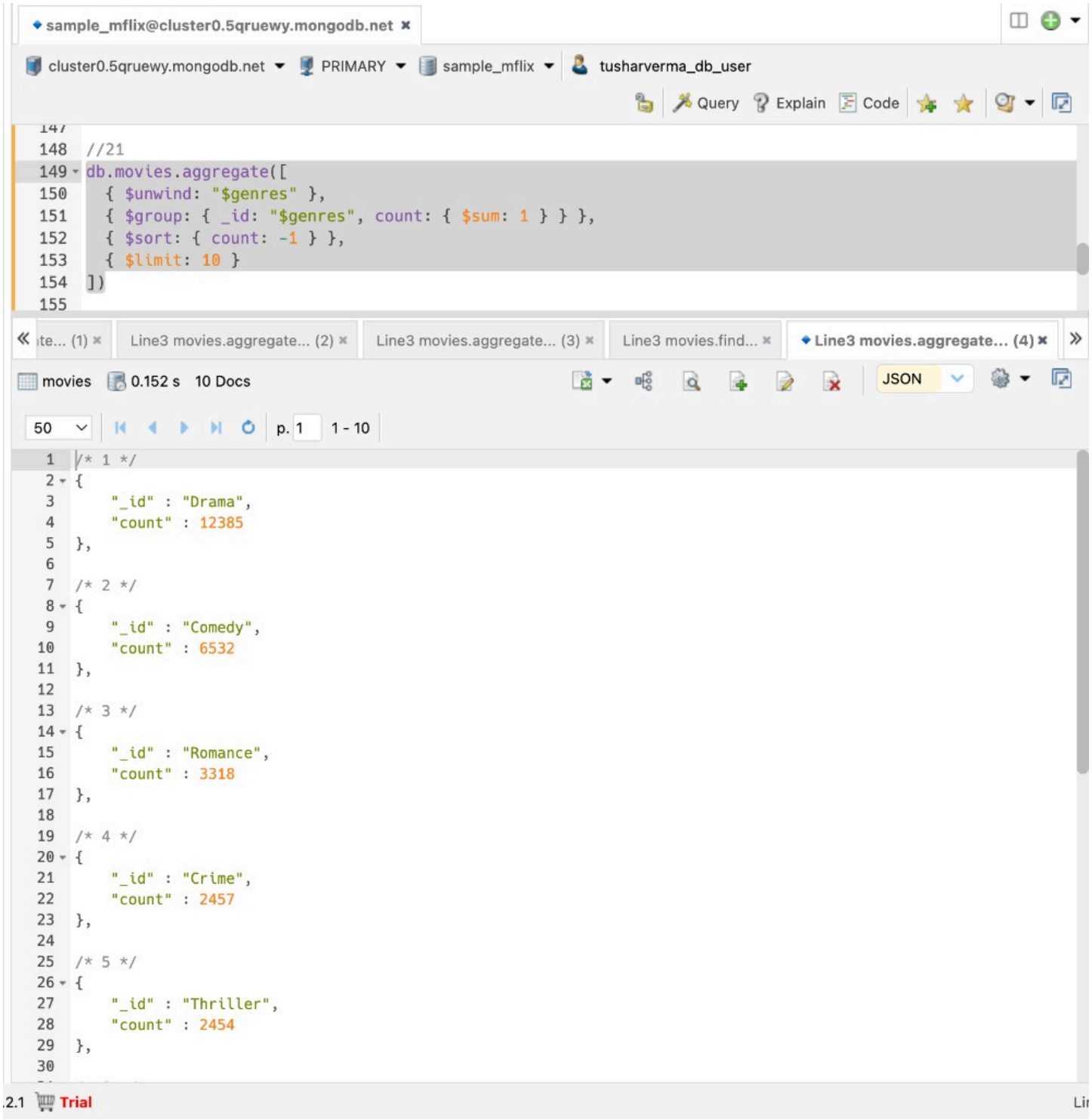
```
db.my_movies_tushar.find({ rated: "PG" })

db.my_movies_tushar.deleteMany({ rated: "PG" })
```

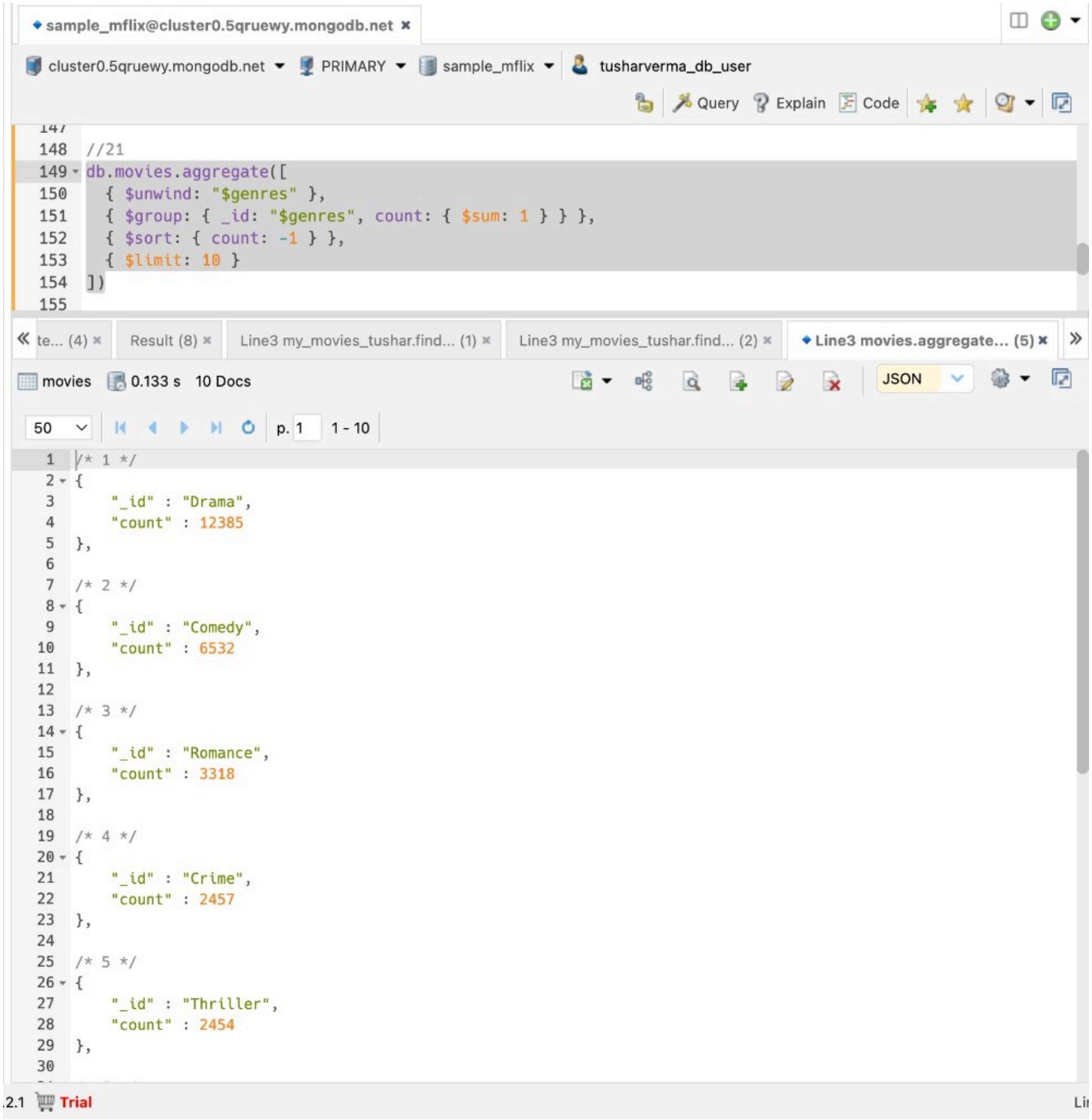
#21 Top 10 genres by movie count

```
db.movies.aggregate([
  { $unwind: "$genres" },
  { $group: { _id: "$genres", count: { $sum: 1 } } },
  { $sort: { count: -1 } },
  { $limit: 10 }
])
```

Screenshot - Task 21 (MongoDB Compass)



Screenshot - Task 21 results (top genres)



#22 Average runtime by rating

```
db.movies.aggregate([
  { $match: { runtime: { $type: "number" } } },
  { $group: { _id: "$rated", avgRuntime: { $avg: "$runtime" } } },
  { $sort: { avgRuntime: -1 } }
])
```

#23 Lookup comments for The Godfather

```
db.movies.aggregate([
  { $match: { title: "The Godfather" } },
  { $lookup: {
      from: "comments",
      localField: "_id",
      foreignField: "movie_id",
      as: "comments"
  } }
])
```

#24 Task 24 (pending)

No query provided yet.

#25 Task 25 (pending)

No query provided yet.

#26 Task 26 (pending)

No query provided yet.

#27 Directors with more than 20 movies

```
db.movies.aggregate([
  { $unwind: "$directors" },
  { $group: { _id: "$directors", movieCount: { $sum: 1 } } },
  { $match: { movieCount: { $gt: 20 } } },
  { $sort: { movieCount: -1 } }
])
```

#28 Countries with 500+ movies + avg IMDb rating

```
db.movies.aggregate([
  { $unwind: "$countries" },
  { $group: {
      _id: "$countries",
      movieCount: { $sum: 1 },
      avgRating: { $avg: "$imdb.rating" }
  } },
  { $match: { movieCount: { $gte: 500 } } },
  { $sort: { movieCount: -1 } }
])
```